Chemistry 3A – Fall 2025 Tuesday Version**: Section 25810**

Name _____ ID _____

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

_**b**_ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
_**e**_ $PV = nRT$
_**d**_ "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
_**a**_ $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm or 760 torr

(b) What is standard temperature in Celsius (°C)? (1 pt)

0° C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

K = C + 273.15 or 273 = 0 + 273.15 K or 0 + 273 K = 273.15 K or 273 K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of $\mathbf{3.50\ L}$ at room **temperature** ($\mathbf{25^\circ C}$). It is placed in a warm environment until its **volume** expands to $\mathbf{4.25\ L}$ at constant pressure. What is the **new temperature** of the balloon in **Celsius (**˚C**)**? (Are intermediate conversions necessary!?)

1. Convert T_1 to Kelvin:
   $25 + 273 = 298\ \text{K}$
   $25 + 273.15 = 298.15\ K$
2. Formula: $\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T_2 = V_2 \times \frac{T_1}{V_1}$
3. Calculation:
   $T_2 = 4.25\ \text{L} \times \frac{298\ \text{K}}{3.50\ \text{L}} = 361.86\ \text{K}$
   $T_2 = 4.25\ \text{L} \times \frac{298.15\ \text{K}}{3.50\ \text{L}} = 362.04\ \text{K}$
4. Convert back to ˚C: $362 - 273 = 88.9^\circ\text{C}$

Rubric (Total 5 pts):

- **5 pts:** Correct setup, Kelvin conversions, algebra, final unit (˚C), and correct sig figs (106°C).
- **4 pts:** Correct calculation and units, but forgot the final conversion back to ˚C (left as 379 K).
- **3 pts:** Correct formula/algebra, but temperature conversion skipped (used 22°C directly).
- **1–2 pts:** Identified variables (V_1, T_1, V_2) and selected the correct formula, but made calculation or algebraic errors.
- **0 pts:** Blank or incorrect approach

4. Kinetic Molecular Theory (3 pts)

According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

Rubric:

- 3 pts: States clearly that collisions are **completely elastic** and **no energy is lost**.
- 2 pts: Mentions energy is conserved or bounces without slowing down.
- 1 pt: General vague statement about moving fast.
- 0 pts: Incorrect statement.

5. Ideal Gas Law Calculation (5 pts)

Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

• **Formula:** $\boldsymbol{PV = nRT \Longrightarrow P = \frac{nRT}{V}}$ **Calculation:** $\boldsymbol{P = \frac{(0.150\ \text{mol})\left(0.08205 \frac{\text{L·atm}}{\text{mol·K}}\right)(310\ \text{K})}{2.50\ \text{L}} =}$ $\boldsymbol{1.53\ \text{atm}}$	**Rubric (Total 5 pts):** • **5 pts:** Completely correct setup, correct R usage, proper algebraic isolation of P, and correct answer (1.53 atm). • **4 pts:** Minor arithmetic or rounding slip, but the methodology is clean. • **3 pts:** Correct formula and inputs, but algebraic rearrangement error (e.g., multiplied by volume instead of dividing). • **1–2 pts:** Identified n, V, T values properly, but stalled or used the wrong gas constant. • **0 pts:** Blank.

6. Partial Pressures and Mole Fraction (8 pts)

A gaseous mixture in a sealed container consists of $\boldsymbol{0.400}$ moles of nitrogen gas (N_2) and $\boldsymbol{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\boldsymbol{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n_{tot}}$) of gas in the container? *(1 pt)*

Total Moles: $n_{tot} = 0.400\ \text{mol} + 0.100\ \text{mol} = \boldsymbol{0.500}\ \text{mol}$

(b) Determine the mole fraction ($\boldsymbol{\chi}$) of oxygen gas (O_2). *(2 pts)*

Mole Fraction: $\chi_{Ne} = \frac{n_{Ne}}{n_{tot}} = \frac{0.100\ \text{mol}}{0.500\ \text{mol}} = \boldsymbol{0.200}$

(c) Calculate the partial pressure ($\boldsymbol{P_{O_2}}$) of oxygen gas in the mixture. *(3 pts)*

Partial Pressure: $P_{Ne} = \chi_{Ne} \times P_{tot} = 0.200 \times 1.25\ \text{atm} = \boldsymbol{0.250}\ \text{atm}$

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

The mole fraction **remains unchanged** (0.200). *Explanation:* Mole fraction is solely determined by the ratio of moles of that specific component to the total moles, independent of changes in total pressure or volume.

Rubric Summary: 8 pts total: Directly checks their grasp of Dalton's proportional partial pressure relationship ($P_A = \chi_A P_{tot}$) and mole fractions without requiring $PV = nRT$ manipulation. Partial credit is assigned step-by-step so a calculation error in part (b) doesn't unfairly cascade through the rest of the problem.